Smart Clinic AI Assistant

Project Proposal

Initiative: DEPI — Digital Egypt Pioneers Initiative **Track:** AI & Data Science **Team Leader:** Ahmed Khaled Abdelfattah

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1. Project Overview

Smart Clinic AI Assistant is an n8n-based automation system that manages a clinic's communication with patients and streamlines internal operations. The system integrates a Telegram chatbot, Google Calendar for scheduling, and Google Sheets as the data store, for a single-doctor clinic in the MVP. It reduces manual work for clinic staff, improves the patient experience with 24/7 availability, and gives the doctor(s) an accurate, automated view of daily clinic performance.

2. Project Objectives

1. **Automated Patient Inquiries** — Answer recurring patient questions (location, fees, working hours) via Telegram, around the clock.
2. **Appointment Booking** — Let patients self-book appointments with real-time availability checks against the calendar.
3. **Automated Reminders** — Send an interactive reminder 24 hours before each appointment (Confirm / Reschedule / Cancel) to reduce no-shows.
4. **Financial Reporting** — Generate an automated daily report for the doctor(s) with total revenue and case count, calculated programmatically (not by the LLM) to guarantee accuracy.
5. **Patient Satisfaction Analysis** — Collect post-visit feedback via Telegram, classify it (Positive / Neutral / Negative), and surface it to support service improvement.

3. Project Scope

In scope for the MVP:

- A Telegram-based conversational interface for patients.

- Button-driven menu (Telegram inline keyboard) to route patients between general inquiries and booking requests, with no LLM involved in this step.
- Appointment booking and availability checks against Google Calendar (single doctor, 20-minute slots).
- 24-hour interactive appointment reminders with status tracking.
- A daily, code-calculated financial summary emailed to the doctor(s).
- Post-visit feedback collection and sentiment classification.
- Data storage via Google Sheets (patients, appointments, daily ledger, feedback).

Out of scope for the MVP (candidate future work):

- Migration to a relational database (e.g., PostgreSQL) if data volume or concurrency needs grow.
- WhatsApp integration (Telegram is used for the MVP due to lower setup overhead).
- Handling of non-text patient messages (voice notes, images) beyond a basic fallback reply.
- Multi-branch / multi-clinic support.

4. Target Users

- **Patients** — interact with the system via Telegram to ask questions, book appointments, respond to reminders, and give feedback.
- **Doctor(s)** — receive the daily financial report and benefit from fewer no-shows and better visibility into clinic performance.
- **Clinic secretary/staff** — continue to record payments and attendance in the shared ledger that the system reads from.

5. Proposed Solution

The system is built as a set of n8n workflows orchestrating a Telegram bot (button-driven menus), Google Calendar, Google Sheets, Gmail, and an LLM used only for post-visit feedback sentiment classification:

Module	Trigger	Summary
1. Message Handling & Booking	Telegram message	Routes the patient via Telegram inline-button menus (no LLM); answers inquiries with fixed clinic info; collects patient data and books appointments against calendar availability
2. Reminders & Follow-up	Daily/periodic schedule	Sends a 24h-ahead interactive reminder; updates booking status based on patient response (Confirmed / Cancelled / Needs Rescheduling)
3. Daily Financial Report	End-of-day schedule	Reads the day's ledger, computes totals via code (not LLM), and emails a summary to the doctor(s)
4. Patient Feedback & Sentiment	Scheduled after appointment	Requests feedback via Telegram, classifies sentiment, and logs it for review

Data store: Google Sheets is used for the MVP (Bookings, Daily Ledger, Feedback) for fast setup and team accessibility, with a logical data model designed so it can be migrated to a relational database later without a redesign.

6. Assumptions & Open Items

- Clinic scope confirmed: single doctor, 20-minute appointment slots. Cancellation/reschedule policy defaults to 3 hours before the appointment, pending confirmation with the client/instructor.
- Deployment target (working prototype vs. production-ready system) to be confirmed.
- Telegram is assumed as the patient-facing channel for the MVP; WhatsApp is a possible future direction.

Project Plan

Timeline (Milestones)

Aligned to the official DEPI submission dates for this track.

Milestone	Deliverables Due	Date
M1 - Planning & Management	Project Proposal, Project Plan, Task Assignment, Risk Assessment, KPIs	16 Oct 2026
M2 - Requirements & Literature Review	Stakeholder Analysis, User Stories, Functional/Non-functional Requirements, Literature Review	16 Oct 2026
M3 - System Analysis & Design	Use Case/Sequence/DFD diagrams, ER Diagram & Schema, Wireframes, Architecture	6 Nov 2026
M4 - Implementation	Working n8n workflows, source code/export on GitHub, README	30 Nov 2026
M5 - Final Presentation & Reports	User Manual, Technical Documentation, Presentation, Infographic	4 Dec 2026

Gantt Overview (by week)

Phase	Weeks	Focus
Requirements & Design	Weeks 1-3 (to 6 Nov)	Confirm clinic requirements, finalize scope, produce all System Analysis & Design artifacts
Core Build	Weeks 4-6	Module 1 (Telegram Assistant) and Module 2 (Booking)
Extended Build	Weeks 6-7	Module 3 (Reminders) and Module 4 (Financial Report)
Final Build & QA	Week 7 (to 30 Nov)	Module 5 (Feedback), end-to-end testing, bug fixes

Phase	Weeks	Focus
Wrap-up	Week 8 (to 4 Dec)	Documentation, presentation, infographic

Resource Allocation

- Team Leader (self): planning, architecture decisions, documentation review, integration oversight.
- Member 2: Telegram bot + LLM intent/response logic.
- Member 3: n8n workflow automation (Calendar, Sheets, Gmail integrations).
- Members 4 & 5: testing, bug tracking, documentation support, presentation.

Pending confirmation of each member's actual skills and availability - see Task Assignment below.

Task Assignment & Roles

Member	Primary Responsibility	Key Deliverables
Ahmed Khaled Abdelfattah (Team Lead)	Planning, architecture, documentation, integration review	All Planning & Management docs, final architecture sign-off
Ahmed Hossam Abdullah	n8n automation & integrations (Modules 2-3)	Router/classifier, inquiry & sentiment prompts
Mohamed Alaa Ibrahim	Telegram bot & LLM logic (Modules 1, 4-5)	Calendar/Sheets booking and reminder workflows
Ziad Ahmed Fathy	Testing & QA	Test cases, bug reports
Ahmed Mossa Anwar	Documentation & presentation support	User manual, slides, infographic

This split is a starting proposal and will be adjusted once each member confirms their technical background and available hours.

Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Low team member engagement	Medium	High	Confirm availability early; redistribute tasks among active members if needed
Delays building n8n workflows	Medium	Medium	Start with a minimal working version per module before adding features
Integration issues between services	Medium	Medium	Test each integration in isolation before connecting modules
Double-booked appointments	Low	High	Availability check always goes through Calendar/Code Node, never the LLM alone
Financial report errors	Low	High	All calculations done in a Code Node, not by the LLM
Inaccurate AI responses	Medium	Medium	Keep the LLM's role limited to language understanding/generation; business logic stays in code
Patient data protection	Medium	High	Limit data collected to what's needed; restrict booking actions to the requesting patient's own chat_id
Missing the final deadline	Low	High	Follow the milestone schedule above; flag slippage early to the team

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KPIs (Key Performance Indicators)

KPI	Target	How It's Measured
Response Time	Reply within a few seconds of a patient message	Timestamp difference, Telegram message to bot reply
Inquiry Resolution Rate	High share of inquiries answered without human handoff	Ratio of inquiries answered by Module 1 vs. escalated

KPI	Target	How It's Measured
Booking Success Rate	High share of booking attempts completed successfully	Confirmed bookings / total booking attempts
Reminder Delivery Rate	Reminders sent for all confirmed appointments	Reminders sent / appointments due
Financial Report Accuracy	Report totals match the ledger exactly	Automated total vs. manual spot-check
Feedback Collection Rate	Reasonable share of patients respond to the feedback request	Feedback responses / feedback requests sent
Milestone Completion Rate	All milestones delivered on or near their due date	Milestones completed on time / total milestones

Exact target percentages to be set once the MVP is running and we have baseline numbers from testing.